Tristan Toshiharu Endo

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Education

University of California, San Diego – *B.S. Math–Computer Science*

Relevant Coursework: Real Analysis, Stochastic Processes, Brownian Motion, Quantitative Finance, Combinatorics, Probability Theory, Abstract Algebra

Research Interests

Natural Language Processing & AI Hallucination: Studying the patterns and sentence structures of large language model outputs to identify inconsistencies and reduce the propagation of AI-generated misinformation.

Stochastic Processes & Probability Theory: Distributional properties of fractional Brownian motion, extrema of stochastic processes, and conditional distributions of Brownian bridges and Ornstein-Uhlenbeck processes.

Neural Network Theory & Kernel Methods: Convergence behavior of finite-width networks to their infinite-width Gaussian process limits, conjugate kernel recursions, and spectral properties of activation functions.

Algebraic Geometry & Combinatorics: Geometry of Springer fibers, flag variety structure, symmetric group actions, and combinatorial characterizations of irreducible components.

Quantitative Finance & Time-Series Modeling: ML-driven trading strategy development, financial time-series analysis, and stochastic modeling of asset price dynamics.

Research Experience

Undergraduate Researcher — *Advisor: Alexander Cloninger* *Jan 2026 – Mar 2026*
Neural Network Gaussian Processes & Conjugate Kernels

- Derived closed-form conjugate kernels for infinite-width ReLU and erf networks, confirming alignment between finite-width simulations and theoretical kernel recursion predictions
- Investigated kernel degeneracy, spectral bias, and expressivity as functions of activation choice and weight variance, comparing finite-width networks to their infinite-width Gaussian process limits

Undergraduate Researcher — *Advisor: Raymond O. Chou* *February 2026 – Present*
Springer Fibers (Algebraic Geometry & Combinatorics)

- Investigate the geometry and combinatorics of Springer fibers, analyzing flag variety structure and symmetric group actions to characterize irreducible components

Industry Experience

Quantitative Researcher / AI Engineering Intern *January 2026 – Present*
Tekly Studio — Quantitative Research & AI Engineering Lab *San Francisco, CA*

- Developed and backtested ML-driven trading strategies on financial time series, evaluating performance via Sharpe ratio and drawdown analysis
- Engineered data pipelines for preprocessing, simulation, and model evaluation, enabling rapid iteration across strategy variants

Audio Engineer / Analyst *April 2025 – Present*
Think Twice Legal *San Bruno, CA*

- Optimized Demucs-based source separation models via hyperparameter tuning, improving reconstruction fidelity for forensic audio analysis in legal proceedings
- Built preprocessing and normalization pipelines for consistent model input across varying sample rates and noise conditions

Selected Projects

Stochastic Simulation of Brownian Motion & OU Processes *Sep 2025 – Dec 2025*

- Simulated fractional Brownian motion across varying Hurst parameters to characterize long-range dependence and correlation structure
- Quantified empirical distributions of extrema via large-scale Monte Carlo simulation, validating convergence to theoretical predictions across parameter regimes

Audio Signal Processing & ML Pipeline Development *Apr 2025 – Present*

- Designed FFT-based noise reduction models to isolate target signals from complex audio environments
- Built automated stem separation pipelines using Demucs with normalization workflows for consistent model input across varied audio conditions

Technologies

Languages: C, C++, HTML, Java, Matlab, Python, R, SQL

Areas: Algorithmic Trading, Time-Series Modeling, Stochastic Processes, Audio Source Separation, Statistical Modeling